

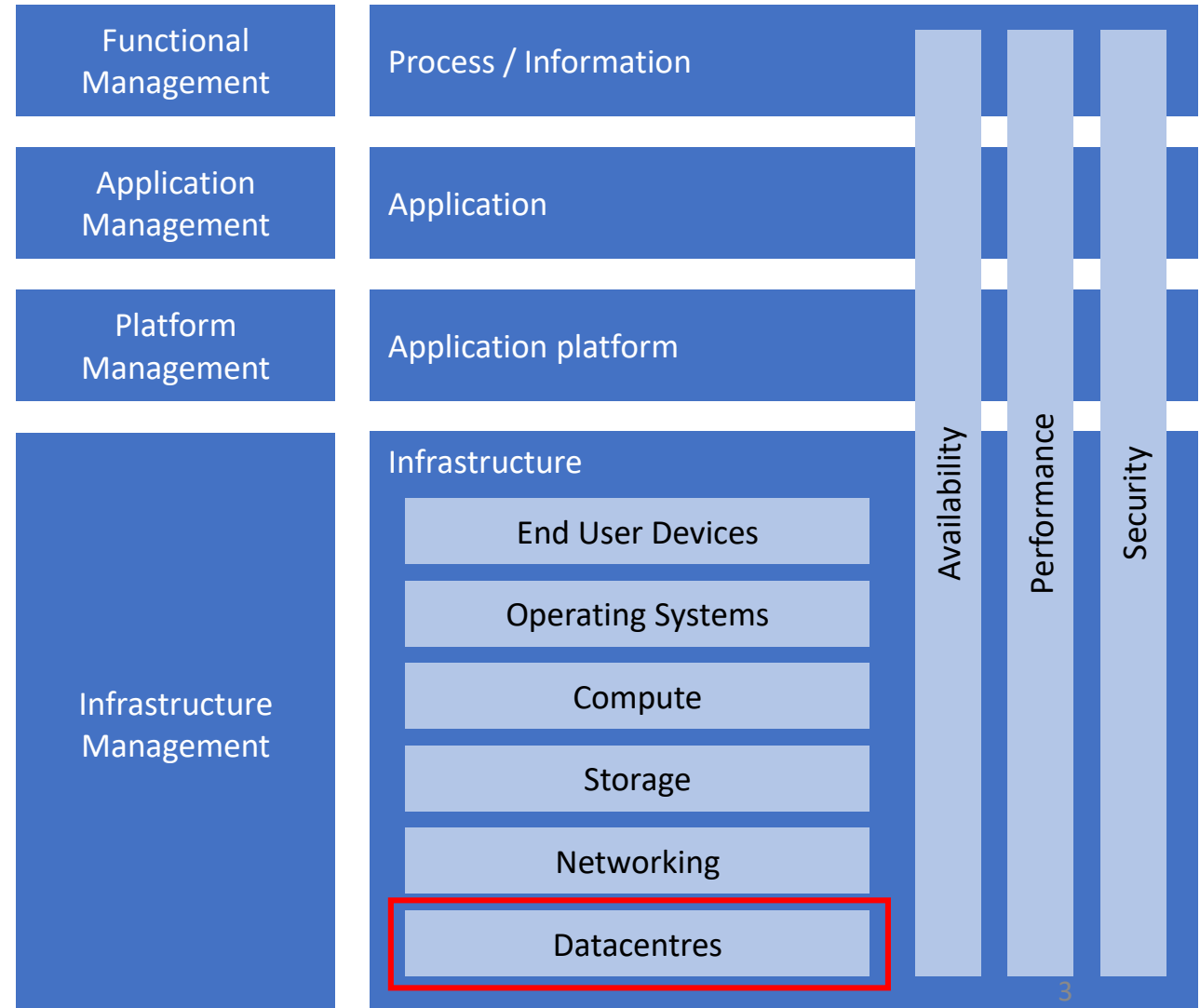
Chapter 3: Datacenter

Objectives

- Explain the functions of a datacenter
- Explain datacenter building blocks
- Explain datacenter
 - Availability
 - Performance
 - Security

Introduction

- Most IT infrastructure hardware, except for end user devices, are hosted in datacenters
- A datacenter provides:
 - Power supply
 - Cooling
 - Fire prevention and detection
 - Equipment racks



History

- Early datacenters (computer rooms) were designed and built for large mainframe systems
- A mainframe, together with its peripheral systems easily filled up a large computer room



Today's datacenters

- Today's datacenters are equipped with standardized 19" racks
 - They house servers, storage devices, and network equipment
 - They are installed in rows forming corridors between them
- Very large datacenters today contain shipping containers packed with thousands of servers each
 - When repairs or upgrades are needed, entire containers are replaced



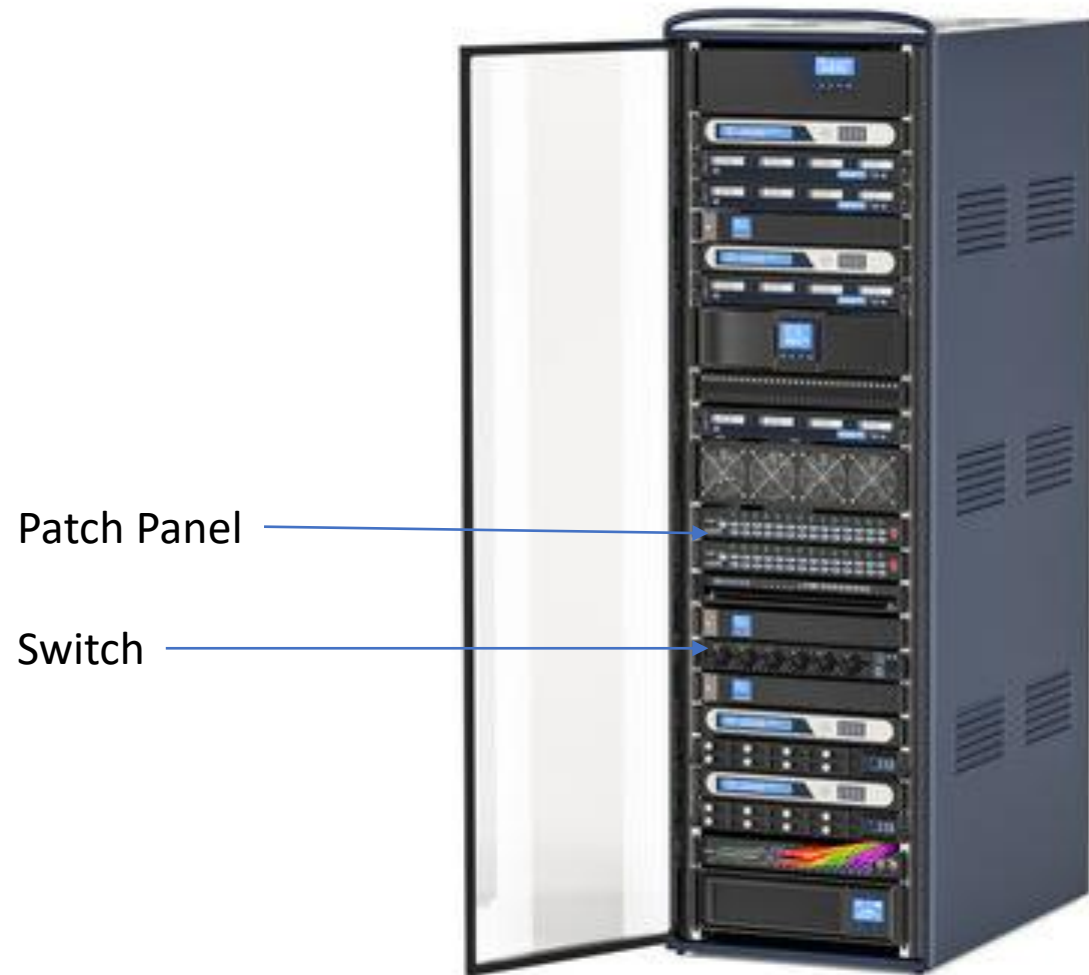


Datacenter building blocks

Datacenter categories

- Typical datacenter categories:
 1. Sub Equipment Room (SER)
 2. Main Equipment Room (MER)
 3. Organization owned datacenter
 4. Multi-tenant datacenter

Sub Equipment Room (SER)



Main Equipment Room (MER)







Amazon data center in Boardman



Cloud datacenters

- These datacenters use an enormous amount of power. E.g. Amazon datacenters house between 50,000 and 80,000 servers, with a power capacity of between 25 and 30 megawatts .

1 MEGAWATT CAN POWER...



1.2 MONTHS OF POWER
FOR AN AVERAGE U.S. HOME



2 REFRIGERATORS
RUN FOR A YEAR



3,600 MILES DRIVEN BY
AN ELECTRIC CAR



2 60-WATT LIGHT BULBS
RUN NONSTOP FOR A YEAR

Datacenter location

- Many variables should be considered to determine where a datacenter could be installed
- Examples:
 - Environment of the datacenter
 - Visibility of the datacenter
 - Utilities available to the datacenter
 - Datacenters located in foreign countries

Distribution of data centres across Malaysia

(completed and under construction)



UPCOMING DATA CENTRE LAUNCHES IN JOHOR IN 2024

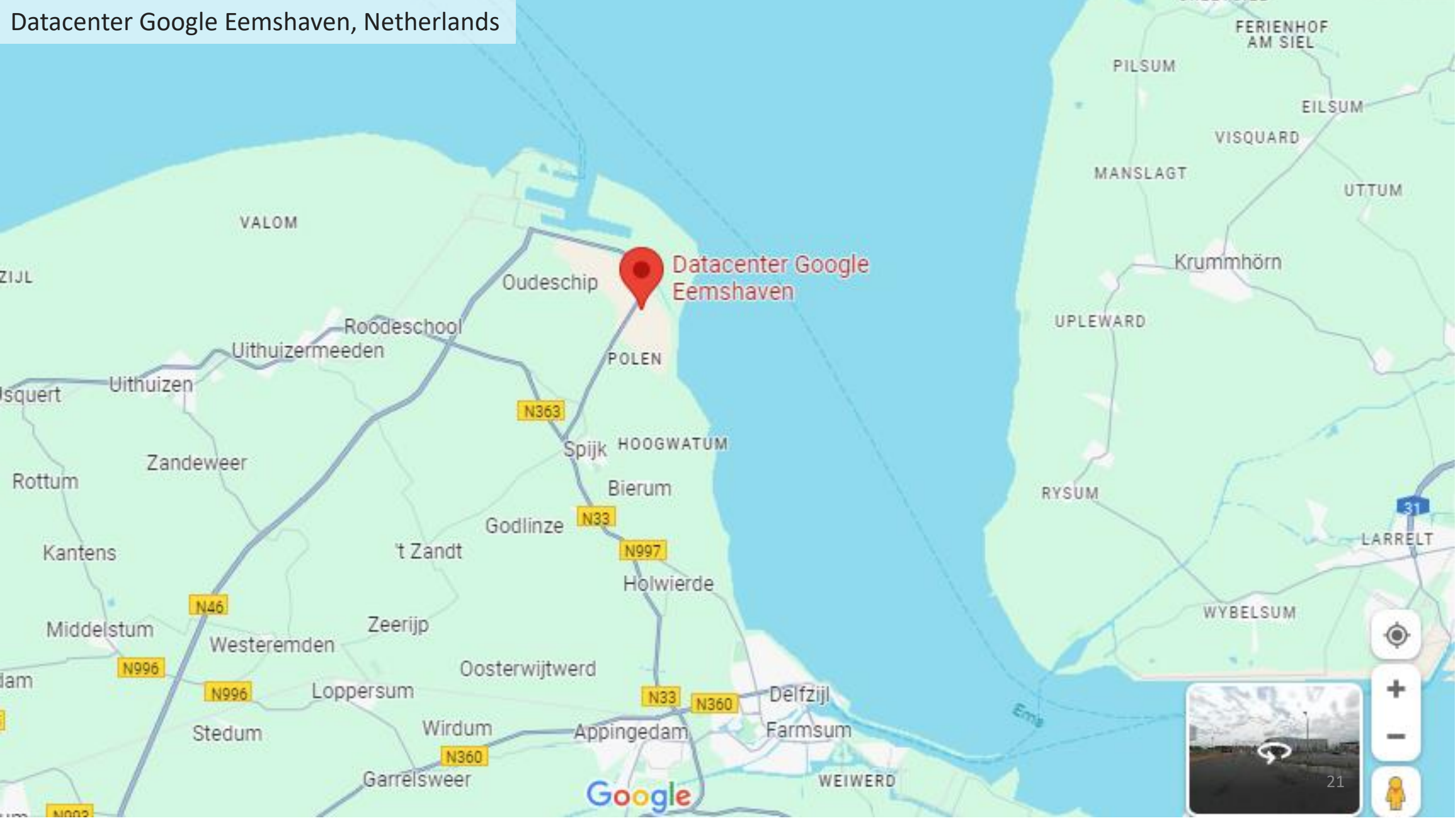
COMPANY / INVESTOR / PARTNER	LOCATION	TOTAL POWER (MW)	SIZE (SQ FT)	ESTIMATED LAUNCH	ESTIMATED COSTS
YTL Power YTL GROUP 	Kulai	500	530k	1Q 2024 (first phase)	RM15 billion
yondr GROUP 	STP	200	1350k	2024	N.A.
 AIRTRUNK	Johor Bahru	150	235k	2024	N.A.
 HOLDINGS	STP	120	38.5k	1H 2024	RM600 million
 HOLDINGS	Nusajaya Tech Park	54	242k	1Q 2024	RM1.38 billion
 EQUINIX	Nusajaya Tech Park	N.A.	21k	1Q 2024	RM176 million

(list is not exhaustive)









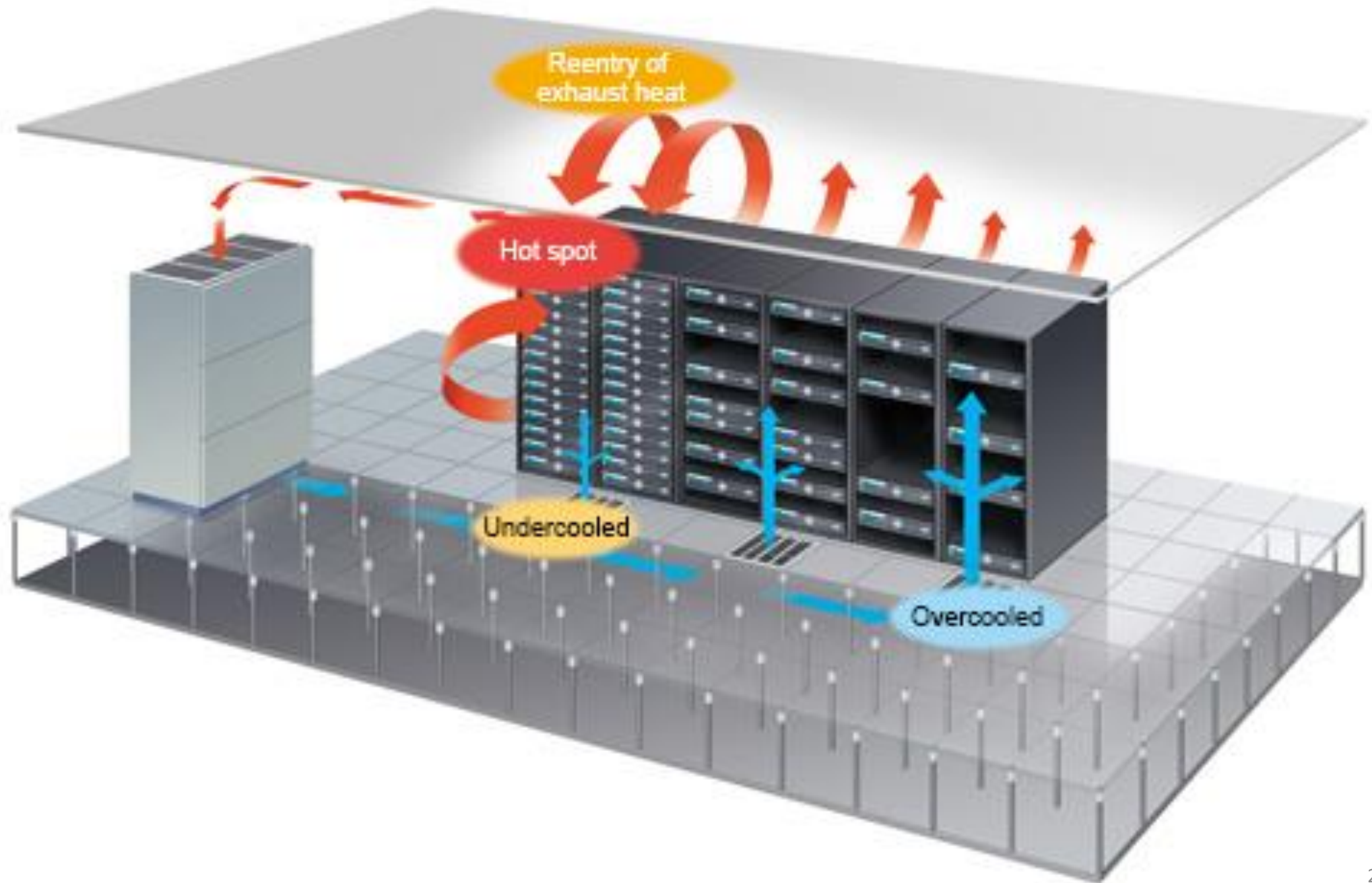
Physical structure

- The physical structure of a datacenter includes components that need special attention:
 - Floors
 - Walls
 - Windows
 - Doors
 - Water and gas pipes

Floors

- In a typical datacenter, the floor must be able to carry 1500 to 2000 kg/m²
 - One fully filled 19" computer rack weighs up to 700 kg
 - The footprint of a rack is about 60x100 cm, leading to a floor load of 1166 kg/m²
 - In office buildings typically the floor can carry approximately 500 kg/m²





Floors

- Raised floors consist of a metal framework carrying removable floor tiles
- Tiles are usually 60×60 cm
- Tiles can be lifted individually to reach cables installed under the raised floor
- Vents provide cool air flow to the racks placed on the floor
- Under the raised floor, data and power cables are installed
 - As alternative, overhead cable trays can be used



Overhead cable trays

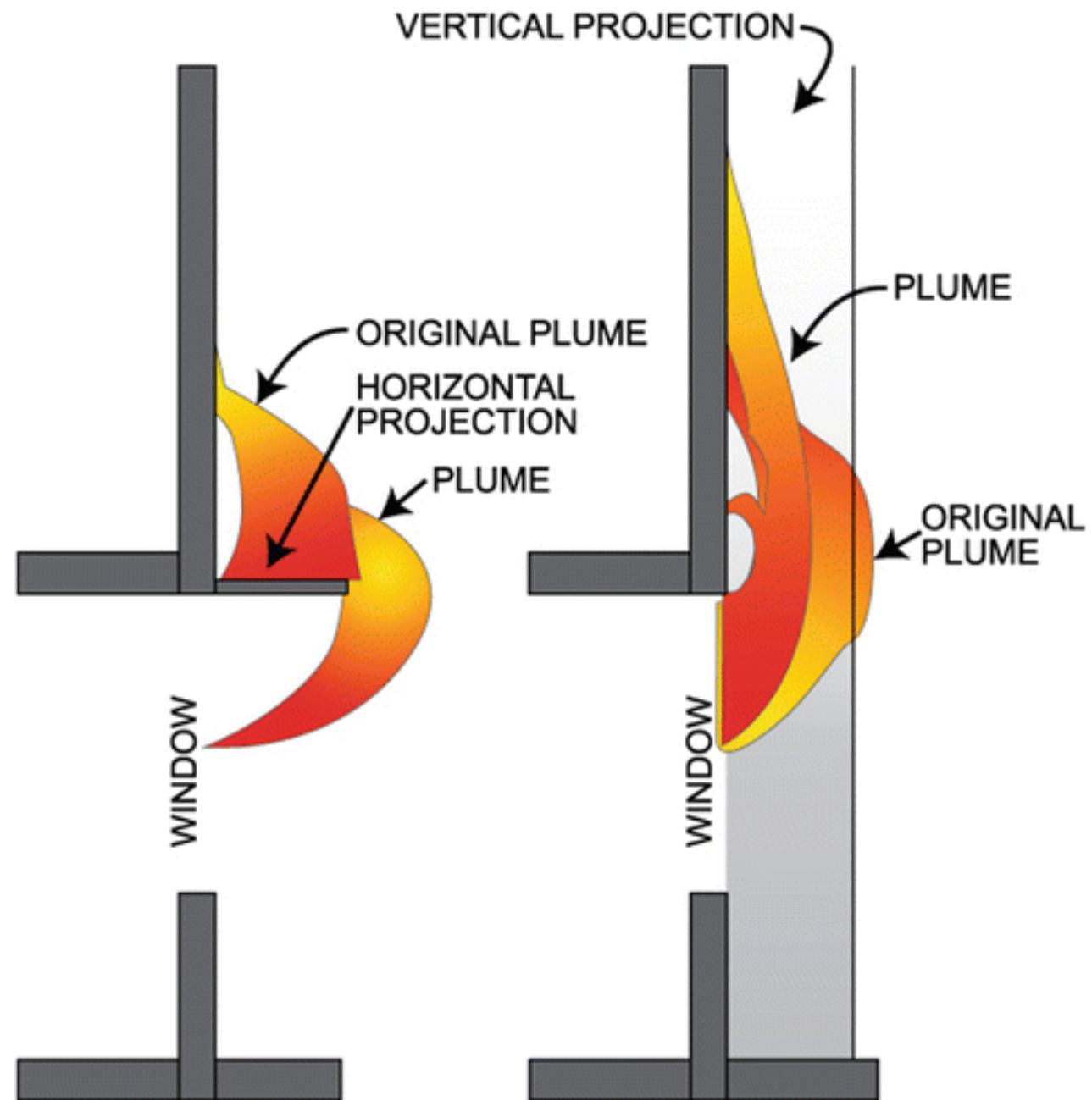


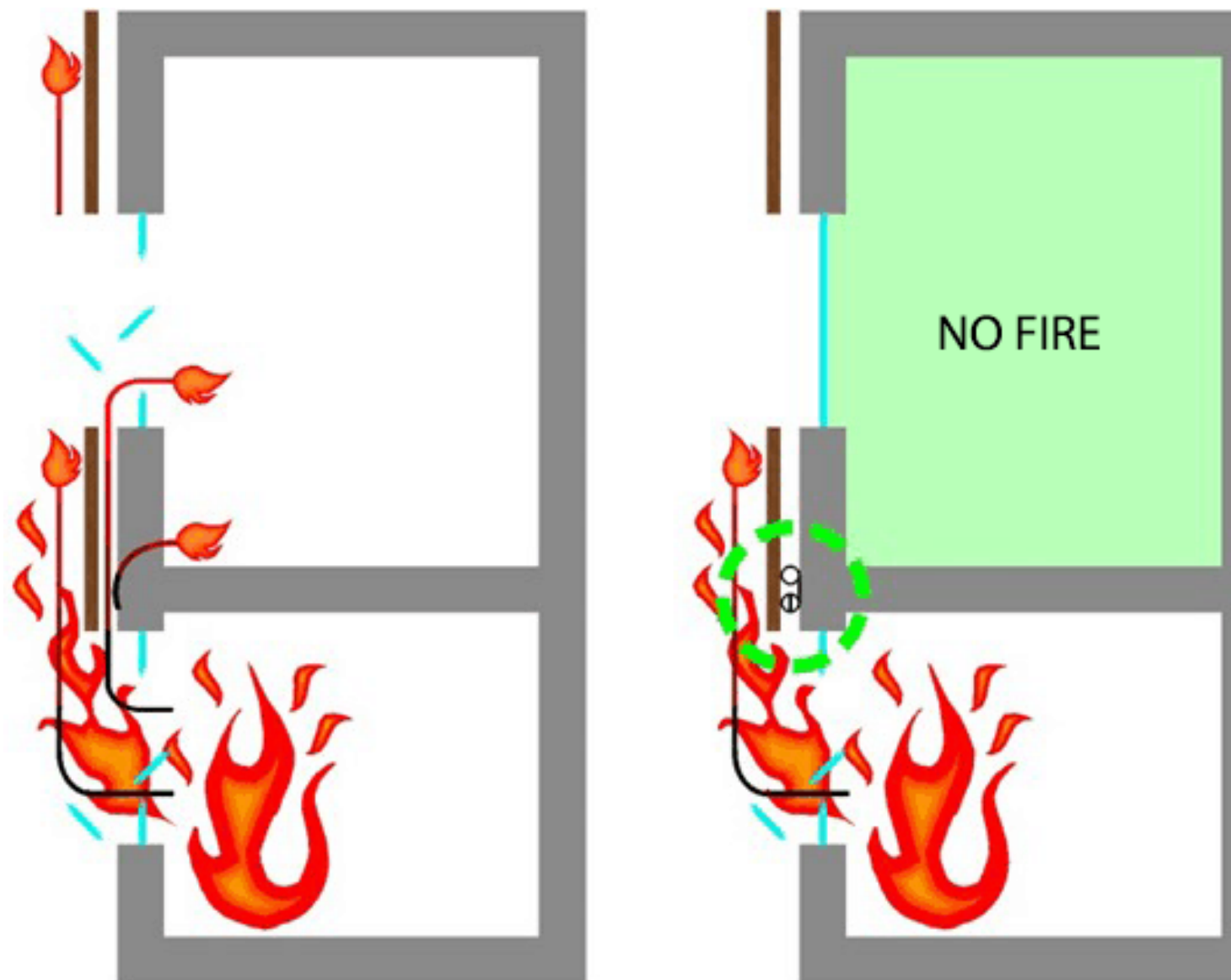
Floors

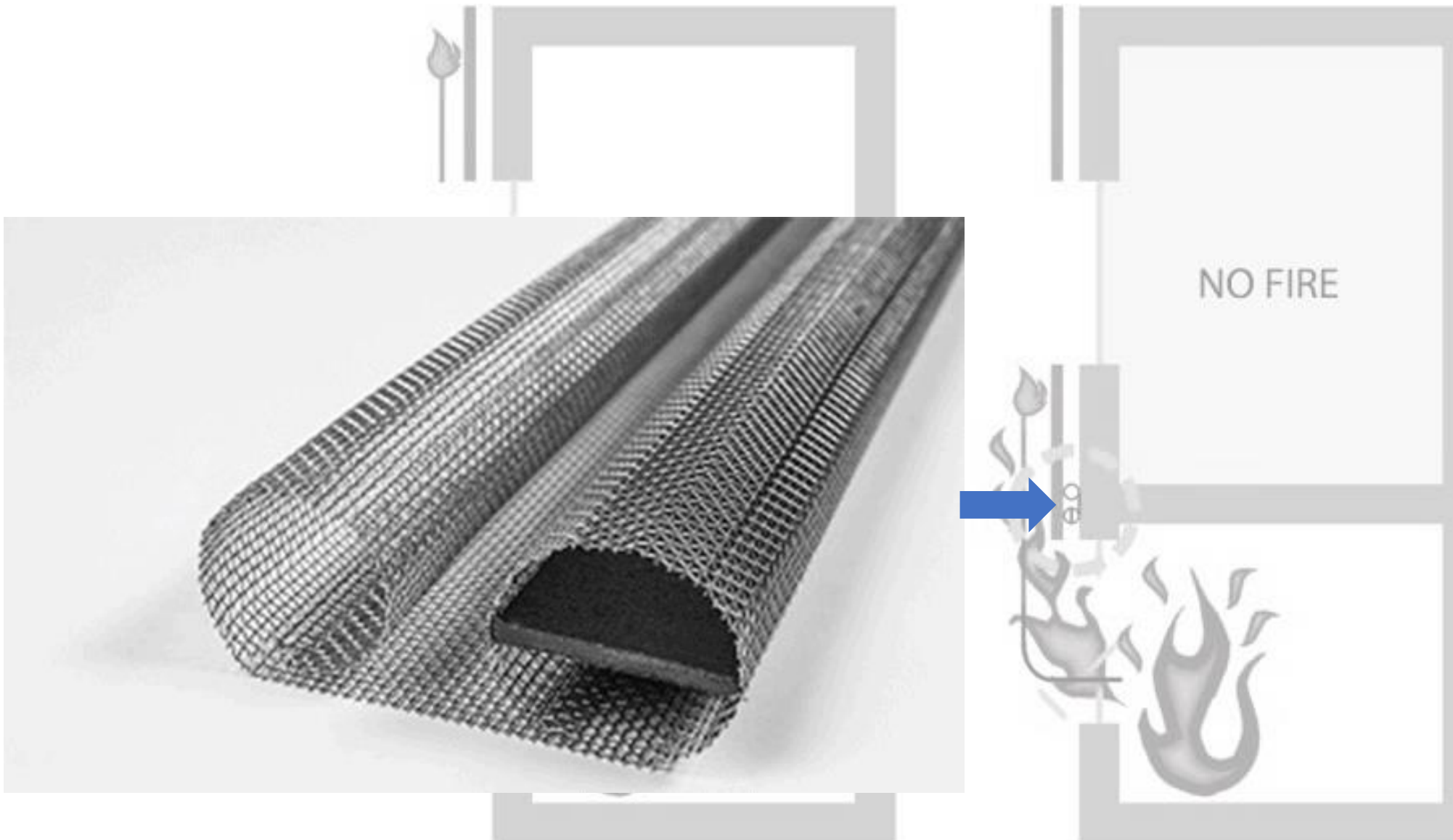
- Raised floors have the following disadvantages:
 - They are **expensive**
 - The total available **height** in the datacenter is **decreased**
 - The **maximum floor load** is **limited**
 - Doors and equipment loading slopes are hard to install due to the difference in floor height
 - Under the raised floor, a **fire** could easily **spread** through the entire datacenter

Walls, windows, and doors

- Walls should reach from the floor to the building's ceiling
 - Because of fire safety and physical intrusion prevention
 - Adequate fire rating is needed to serve as a physical firewall
- Windows are not desirable in a datacenter. If there are windows, they must be:
 - Translucent
 - Shatterproof
 - Impossible to open
- Doors should be large enough to have equipment brought in and must resist forced entry







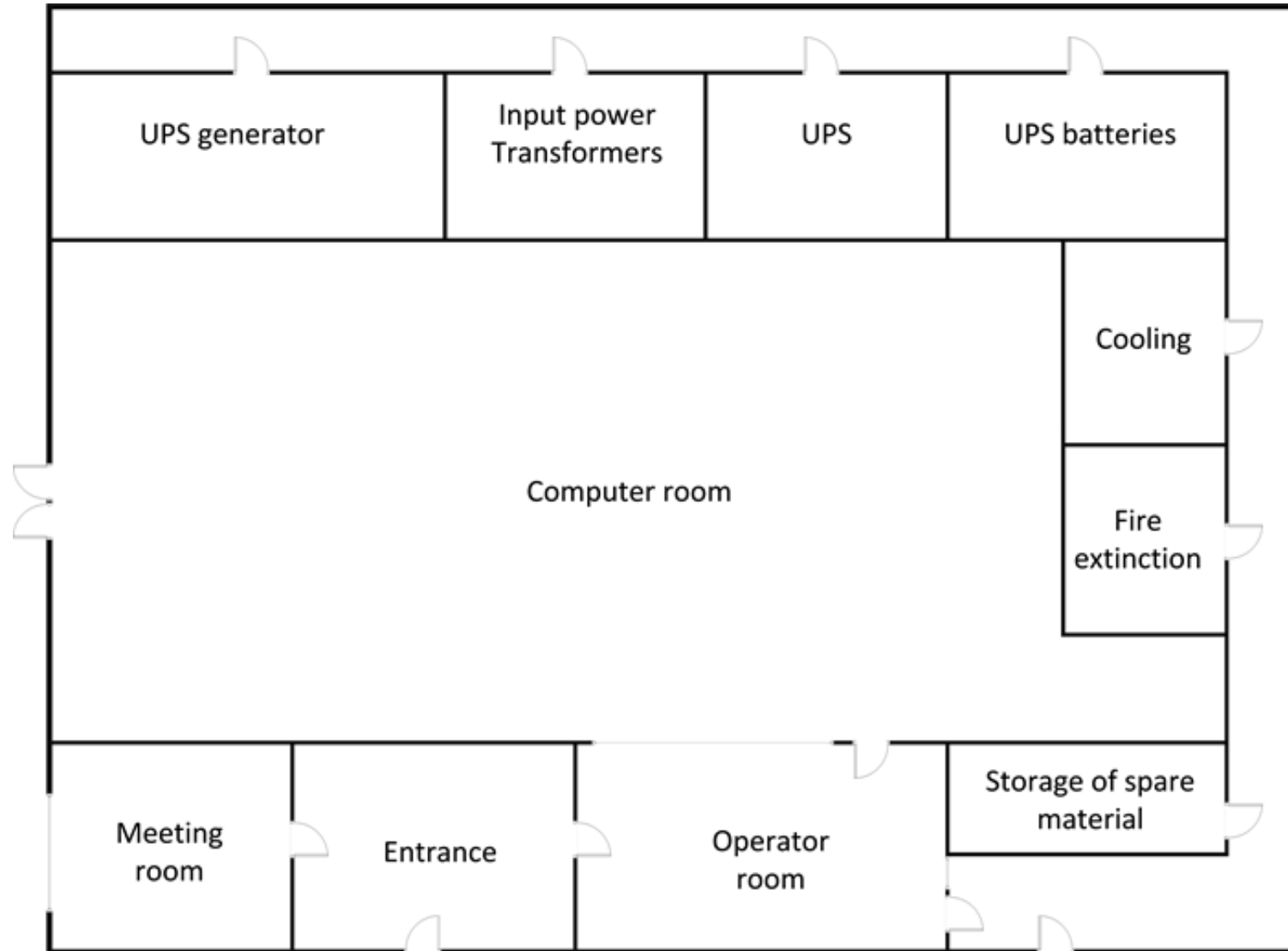
Source: <https://securonorway.com/how-fire-spreads-in-the-facade/>

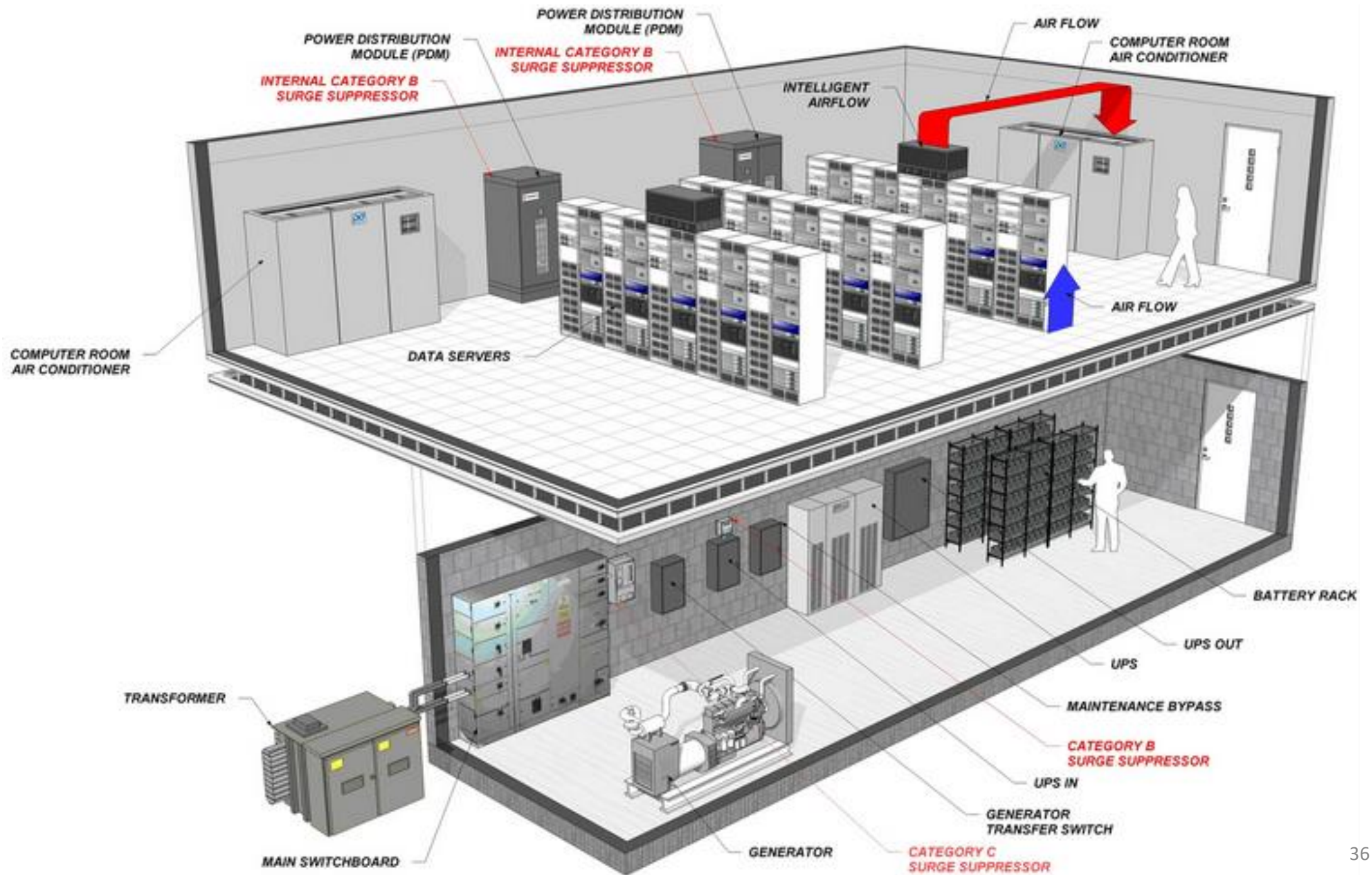


Water and gas pipes

- Water or gas pipes may have been installed under the floor, in the walls, or above the ceiling of the datacenter
- Leakage from water pipes in the ceiling of a datacenter could lead to damage of equipment
- Datacenter operators should know where the shutoff valves are

Datacenter layout





Power supply

- Power drawn by datacenters:
 - A few kilowatts (kW) for one rack of servers
 - Dozens of megawatts (MW) for large facilities
- The amount of power available in a datacenter is expressed as the number of kilowatts per m², known as power density
- Normal-density datacentre: between 2 to 6 kW/m²
- High-density datacentre: between 10 and 20 kW/m²

Power supply

- Example:
 - If a HP DL380 server uses 250W power
 - A rack could physically hold 40 servers
 - Available energy density is 2 to 6 kW/m²
 - Only between 8 and 24 servers can be placed in the rack
- In practice most **server racks cannot be fully equipped**
- A high-density datacentre allows racks to be filled with approximately 40 to 80 servers



Uninterruptable Power Supply (UPS)

- Power issues can occur in the utility power supply leading to downtime and damage to equipment
- Uninterruptable Power Supply (UPS):
 - Is independent of the utility power supply
 - Provides high quality electrical power
- A UPS installation consists of:
 - Filters
 - A diesel power generator
 - A set of batteries or a flywheel system

Uninterruptible Power Supply (UPS)



Small UPS



Datacentre UPS

Diesel power generator



Uninterruptable Power Supply (UPS)

- Battery powered UPSs:
 - Standby UPS systems (also known as off-line systems) used in small setups (a few workstations or servers)
- Line interactive UPS systems
 - Use a transformer between the utility power and the IT equipment
 - Works as a filter for many of the power issues
- Double conversion UPS systems
 - Convert the AC utility power to DC power and then back to high quality AC power

Power distribution

- A **power distribution unit (PDU)** is a device with **multiple power outlets** that **distributes power** to equipment located in the datacenter



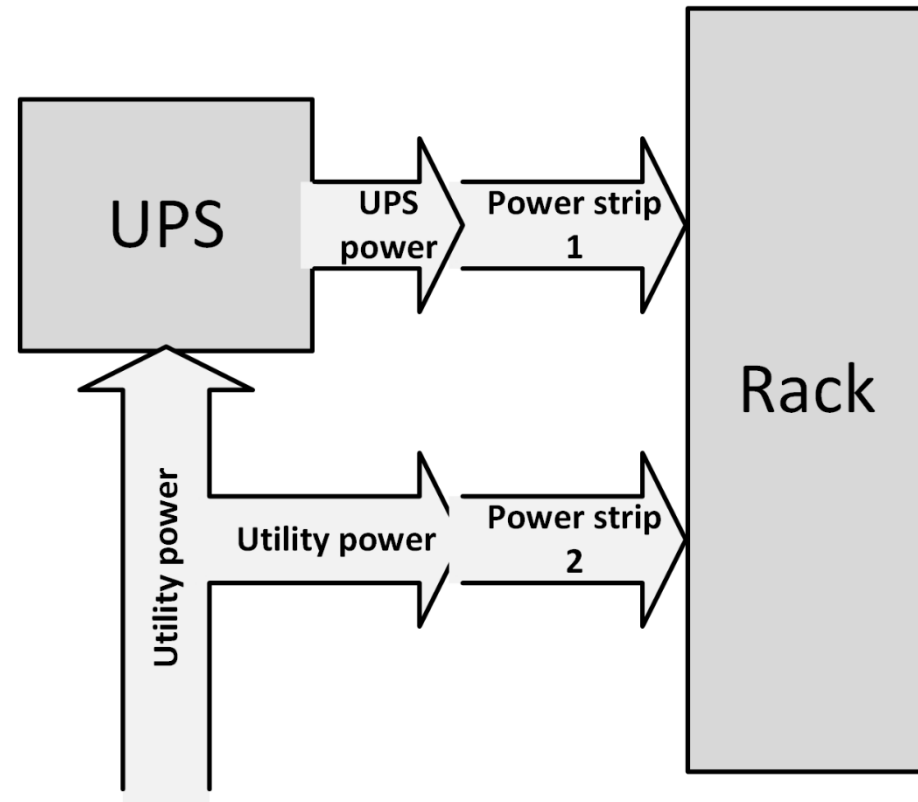
- Two types of PDUs:
 - Large floor mounted PDUs take main feeds (usually 3 phase power) and distribute it into multiple smaller feeds to computer racks
 - Power Strips that feed equipment in racks

Power distribution

- Most Infrastructure components can be equipped with **two power supplies for redundancy**
- For availability reasons at least two power strips are needed to power equipment in a rack
 - Each feeds one of the two power supplies in the equipment

Uninterruptible Power Supply (UPS)

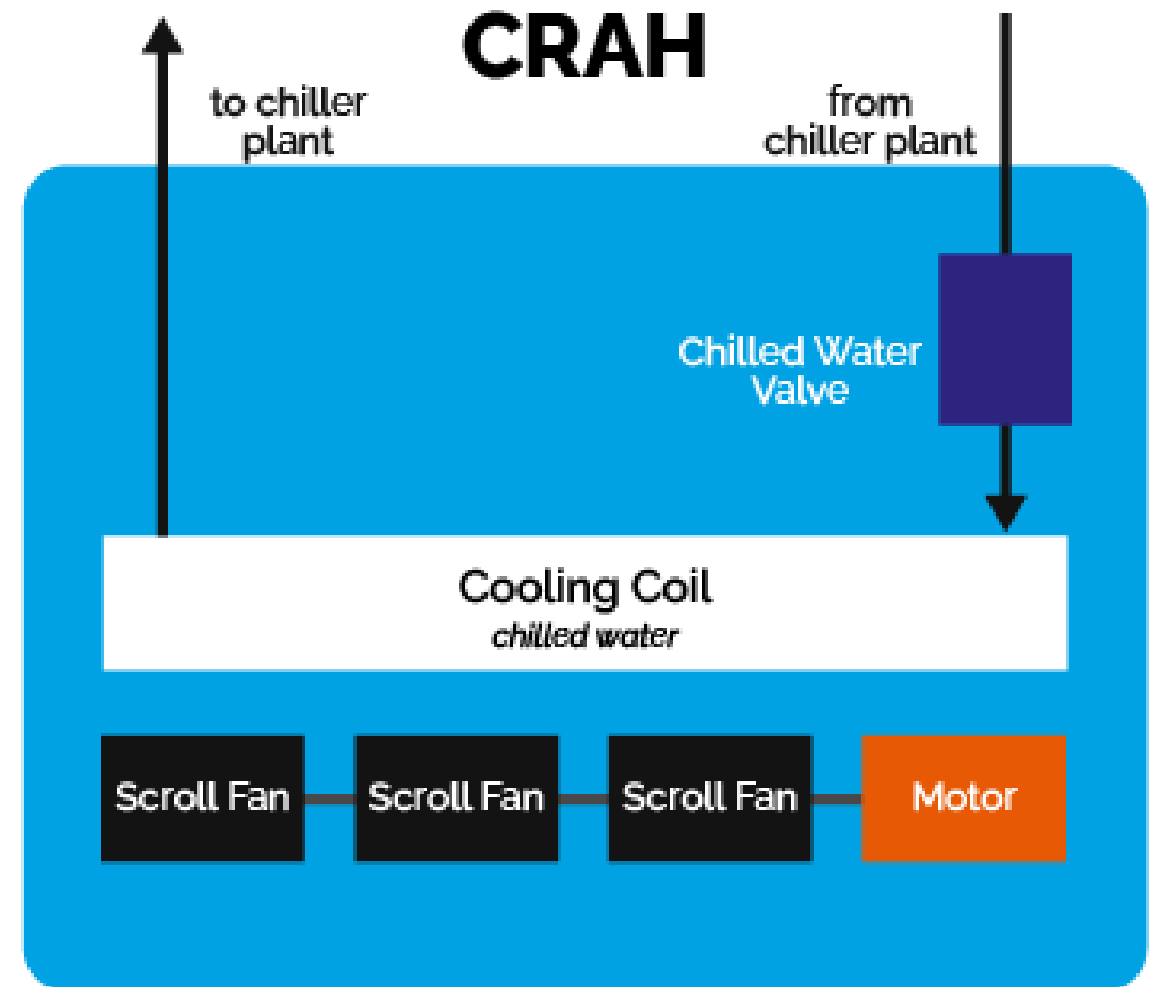
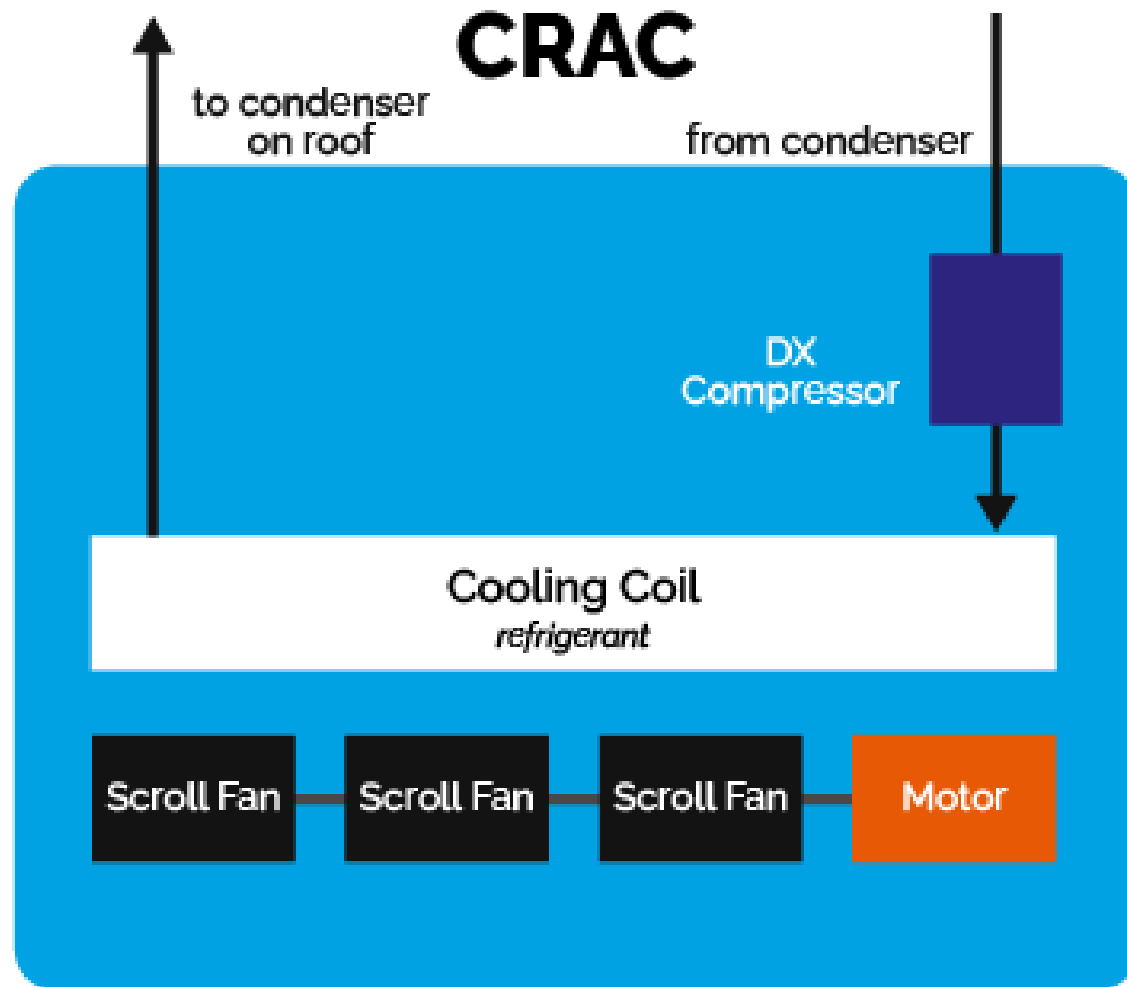
- Two independent power feeds to racks
- UPS should not be a Single Point of Failure



Cooling

- > 90% of all **power** used by IT infrastructure components is converted into **heat**, which has to be dissipated by a cooling system
- Two types of cooling systems:
 - **Computer Room Air Conditioners (CRAC)**: Refrigerant-based units connected to outside condensing units
 - **Computer Room Air Handlers (CRAH)**: Chilled water based and connected to outside chillers. A chiller produces chilled water via a refrigeration process

CRAC VS CRAH



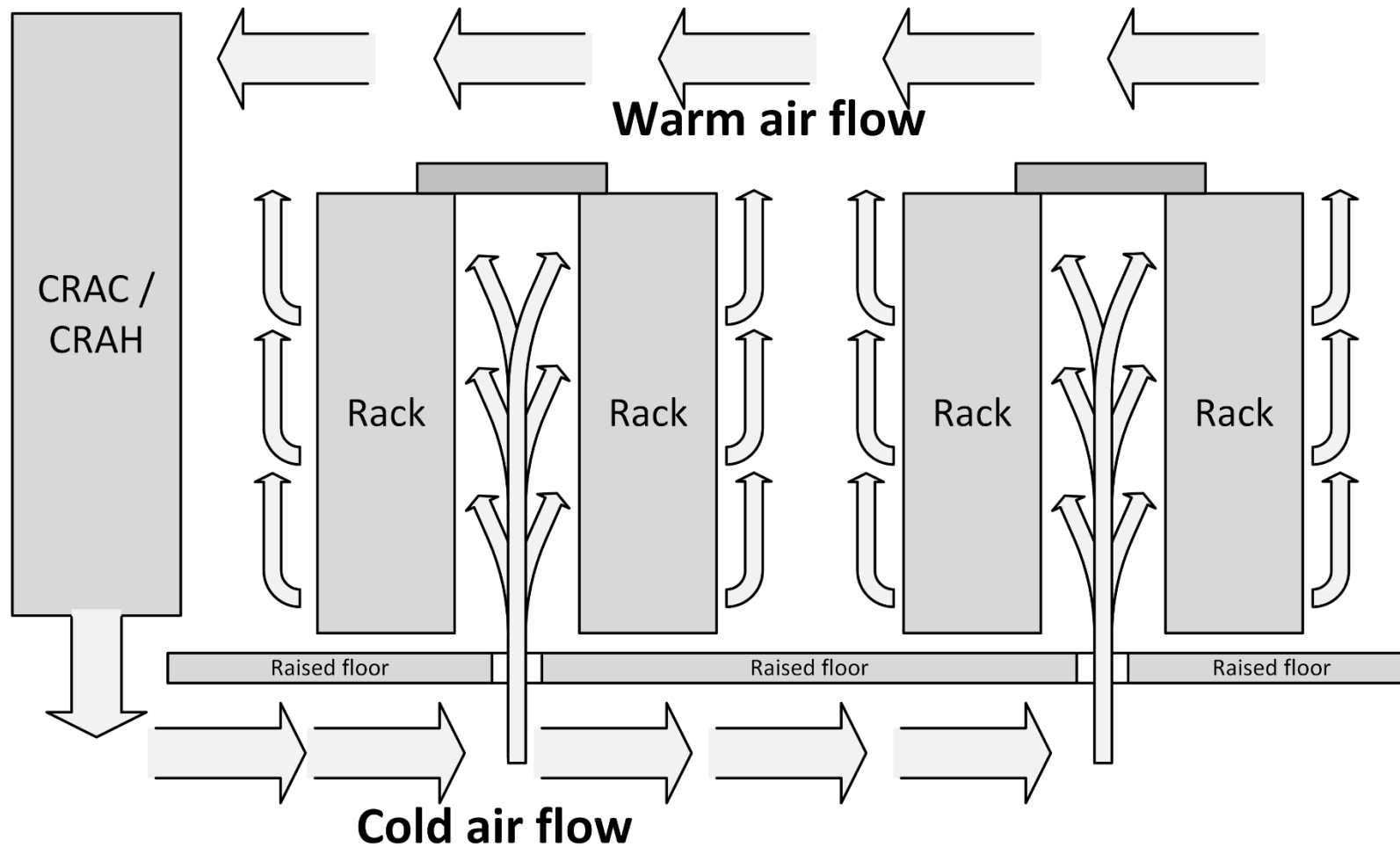
Cooling

- The efficiency of a cooling system is specified in:
 1. **EER - Energy Efficiency Ratio:** The measure of efficiency at maximum air conditioning load. The ratio between output cooling in BTU per hour and the electric energy input in Watts at a given operating point
 2. **SEER - Seasonal Energy Efficiency Ratio:** Same as EER, but seasonal data is used for the measurement. The time of year the cooling system is used most (typically in the summer)
 3. **COP - Coefficient Of Performance:** The ratio between cooling load in kW and the electric energy input in kW. Normal values are between 3 and 10

Operating temperatures

- The air temperature in the datacenter usually ranges from **18 to 27 degrees Celsius**
- Servers **shut themselves down** at an air inlet temperature of **40 degrees Celsius**
- Using higher temperatures saves cooling capacity and power. Raising the temperature with one degree Celsius lowers the cost for cooling by approximately 5%

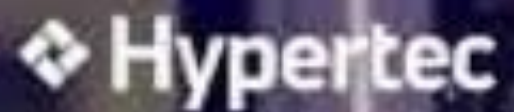
Airflow



Liquid cooling

- Liquid cooling is becoming a standard feature of large datacenters
- It immerses system boards, including all components, in a special non-conductive and non-corrosive fluid
- System boards can be placed much closer together than normal, which increases CPU density in the datacenter





**GET YOUR 360° IMMERSION
COOLING SOLUTION**

Humidity and dust

- The humidity of the air in a datacenter should be between **40% and 60%**
- Minimize the number of dust particles in a datacenter
 - Don't allow visitors in the datacenter
 - People should wear dust-free clothing (like white coats) and protective sleeves around their shoes

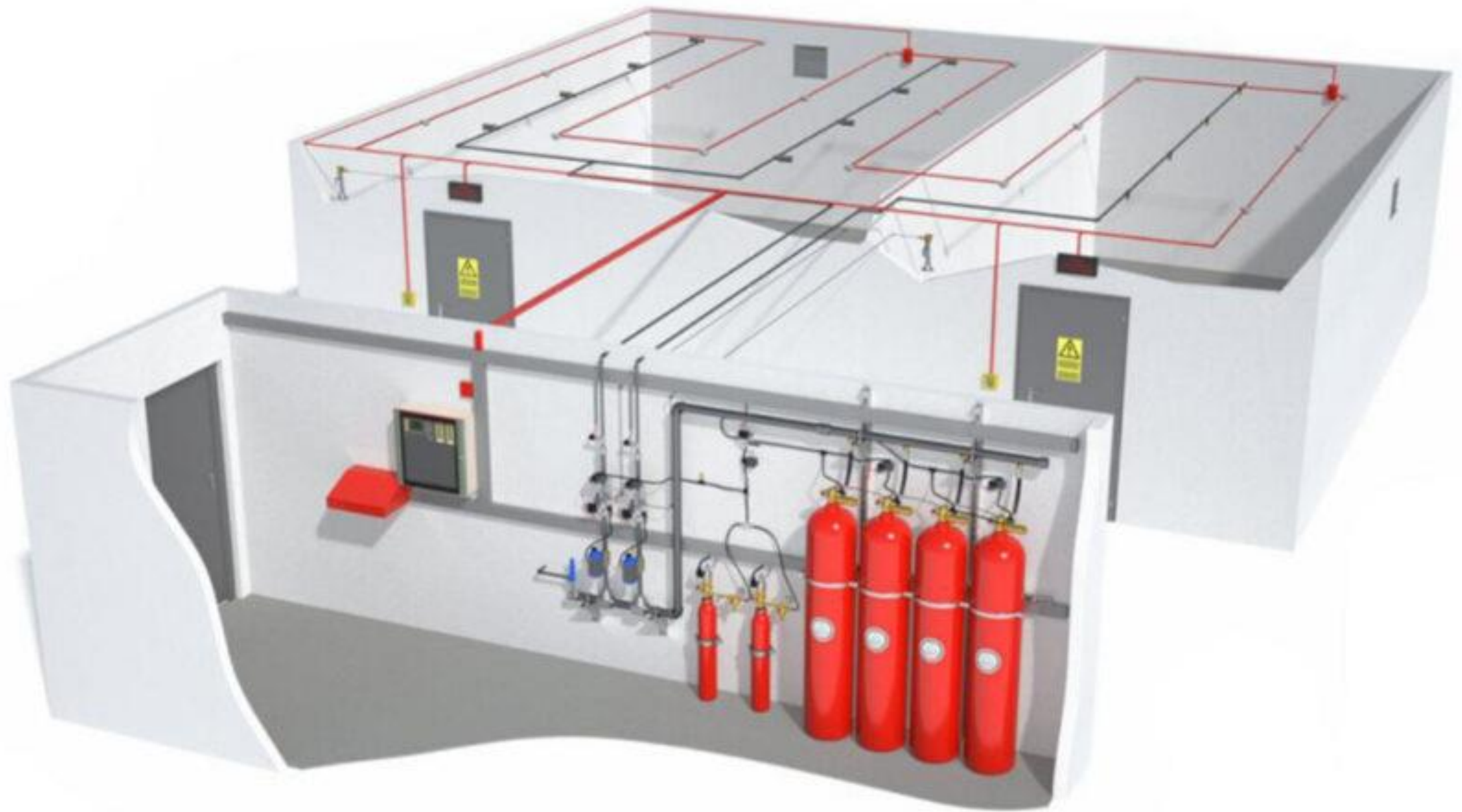
Fire prevention, detection, and suppression

- A short circuit in a cable or defect equipment may cause fire
- Fires can spread around very quickly because of the air flow in the datacenter and the frequent use of raised floors
- Smoke could damage equipment in the datacenter even if a fire starts outside of the datacenter's computer room

Fire prevention, detection, and suppression

- Suppressing fire in a datacenter consists of four levels:
 1. Fire prevention – Avoid a fire
 2. Passive fire protection – Limit the exposure of the fire once it has started
 3. Fire detection systems – Detect smoke and fire
 4. Fire suppression systems – Extinguish the fire once it is detected





CLEAN AGENT
SYSTEM

CABINET TEMP.

88°C

CEILING TEMP.

18°C

SYSTEM PRESSURE

877 kPa

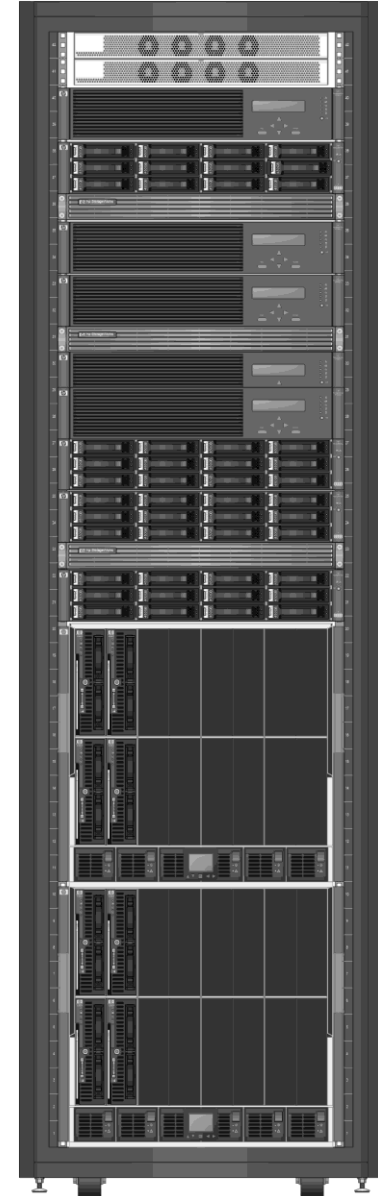
NOVEL EXCHANGE

3m 29s



Equipment racks

- A 19" rack is a standardized metal enclosure to house IT infrastructure components
- The height of a rack is measured in rack unit or 'U'. One U is 44.5 mm
- A typical rack is 42U high



Datacenter energy efficiency

- IT accounts for approximately 1.5% of global electricity use, and 1% of all the world's CO2 emissions
- Energy cost > Server cost
- The Power Usage Effectiveness (PUE) metric measures the power used by the datacenter

Datacenter energy efficiency

- The PUE is calculated by dividing the amount of power used by the datacenter, by the power used to run the IT equipment in it
- Typical PUE value of a datacenter is between 1.1 and 2.0
 - Running a datacenter with a PUE of 1.5 means that for each watt of power used by the IT equipment an extra half watt is used by the rest of the datacenter

Datacenter availability

Availability tiers

Tier	Type	Uptime	Expected Downtime	Measures
1	Basic	99.671%	28.8 hours per year	Single path for power and cooling and few
2	Redundancy Components	99.741%	22 hours per year	Single path for power and cooling and some redundant and backup components
3	Concurrency Maintenance	99.982%	1.6 hours per year	Multiple paths for power and cooling and systems in place to update and maintain it without taking it offline
4	Fault Tolerant	99.995%	26.3 minutes per year	Has redundancy for every component

Availability tiers

- The tier classification only describes the availability of the datacenter facilities, not the availability of the IT infrastructure components
- A tier 3 datacenter running an IT infrastructure with an availability of 99.990% will have a total availability of
- $0.99982 \times 0.99990 = 0.99972 = 99.972\%$

Redundant datacenters

- Multiple redundant datacenters can be used to increase availability
- Multiple datacenters are a must when higher availability than 99.995% is needed
- Redundant datacenters should be at least 5 km apart

Floor management

- Minimize personnel walking around the server racks
- Floor managers responsibilities:
 - Keeping the datacenter floor tidy
 - Changing backup tapes
 - Providing power connections to the racks
 - Maintaining and testing the fire extinguishing system and UPS systems
 - Etc.

Datacenter performance

Datacenter performance

- The datacenter itself does not provide performance to IT Infrastructures, except for the bandwidth of the internet connectivity and the scalability of the location

Datacenter security

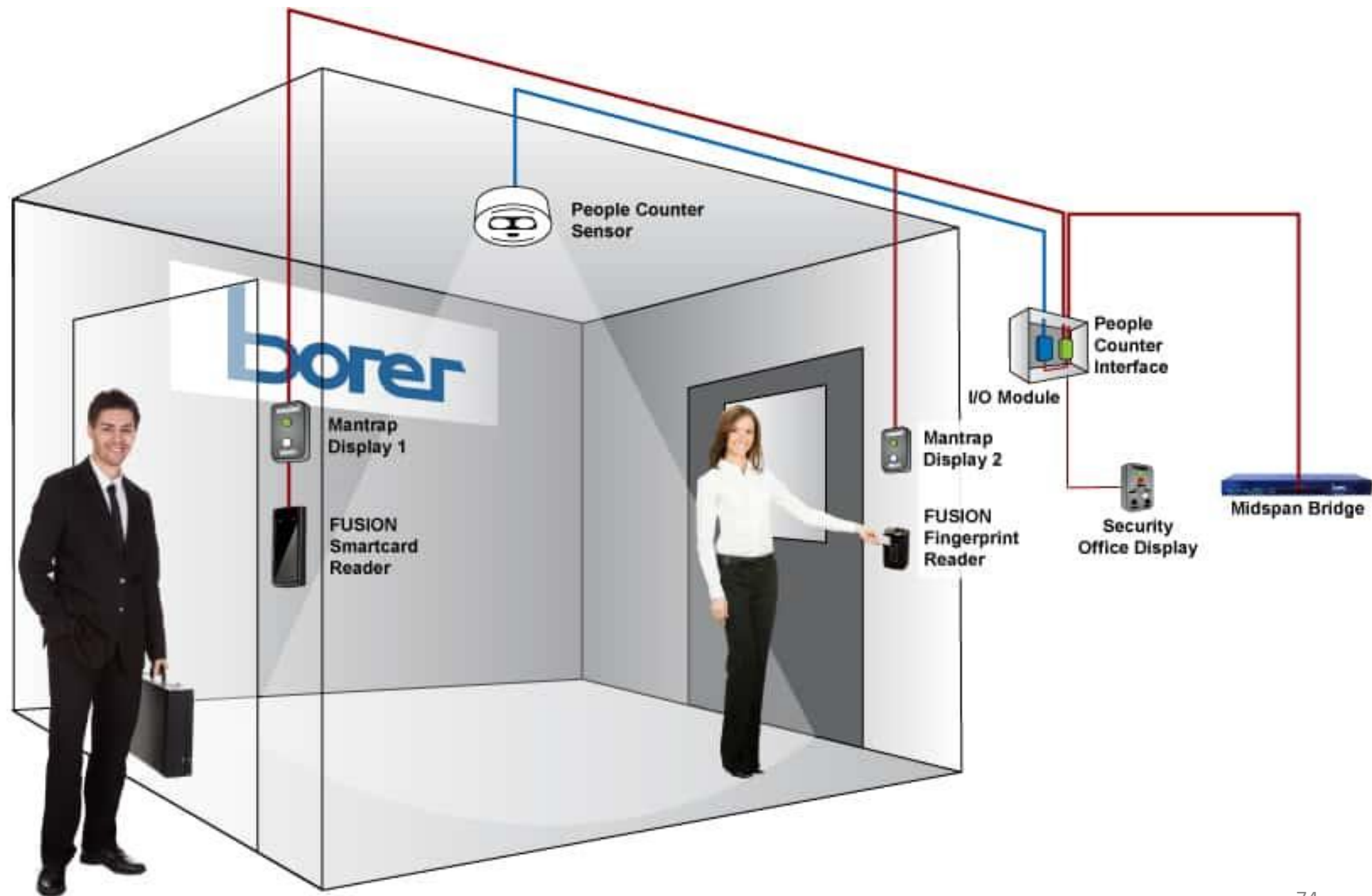
Physical security

- Ensure that equipment is physically safe behind the datacenter doors
- Physical access to the datacenter must be restricted to selected and qualified staff
- An entry registration system should be used
- A log should be maintained containing all staff entering and leaving the datacenter
- Doors must be secured using conventional locks (for instance for dock loading doors) or electronic locks
 - Electronic locks should open only after proper authentication

Physical security

- Entry points can be implemented as:
 - Regular doors
 - Mantraps
 - Staff is routed through a set of double doors that may be monitored by a guard
 - Revolving doors
 - Only one person at a time can enter the datacenter's restricted area
- Entries can be equipped with weighing scales to ensure only one person enters the restricted area

Mantraps





Conclusion

- Explain the functions of a datacenter
- Explain datacenter building blocks
- Explain datacenter
 - Availability
 - Performance
 - Security